

53 GRETH - Ethernet Media Access Controller (MAC) with EDCL support

53.1 Overview

Frontgrade Gaisler's Ethernet Media Access Controller (GRETH) provides an interface between an AMBA-AHB bus and an Ethernet network. It supports 10/100 Mbit speed in both full- and half-duplex. The AMBA interface consists of an APB interface for configuration and control and an AHB master interface which handles the dataflow. The dataflow is handled through DMA channels. There is one DMA engine for the transmitter and one for the receiver. Both share the same AHB master interface. The ethernet interface supports both the MII and RMII interfaces which should be connected to an external PHY. The GRETH also provides access to the MII Management interface which is used to configure the PHY.

Optional hardware support for the Ethernet Debug Communication Link (EDCL) protocol is also provided. This is an UDP/IP based protocol used for remote debugging.

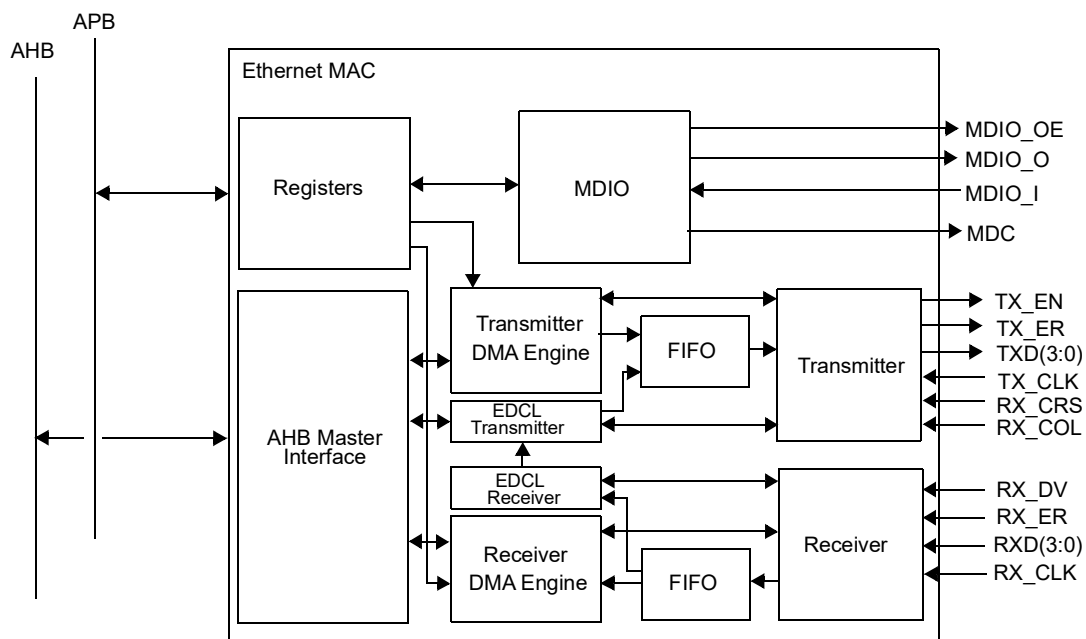


Figure 140. Block diagram of the internal structure of the GRETH.

53.2 Operation

53.2.1 System overview

The GRETH consists of 3 functional units: The DMA channels, MDIO interface and the optional Ethernet Debug Communication Link (EDCL).

The main functionality consists of the DMA channels which are used to transfer data between an AHB bus and an Ethernet network. There is one transmitter DMA channel and one Receiver DMA channel. The operation of the DMA channels is controlled through registers accessible through the APB interface.

The MDIO interface is used for accessing configuration and status registers in one or more PHYs connected to the MAC. The operation of this interface is also controlled through the APB interface.

The optional EDCL provides read and write access to an AHB bus through Ethernet. It uses the UDP, IP, ARP protocols together with a custom application layer protocol to accomplish this. The EDCL contains no user accessible registers and always runs in parallel with the DMA channels.

The Media Independent Interface (MII) is used for communicating with the PHY. There is an Ethernet transmitter which sends all data from the AHB domain on the Ethernet using the MII interface. Correspondingly, there is an Ethernet receiver which stores all data from the Ethernet on the AHB bus. Both of these interfaces use FIFOs when transferring the data streams. The GRETH also supports the RMII which uses a subset of the MII signals.

The EDCL and the DMA channels share the Ethernet receiver and transmitter.

53.2.2 Protocol support

The GRETH is implemented according to IEEE standard 802.3-2002 and IEEE standard 802.3Q-2003. There is no support for the optional control sublayer. This means that packets with type 0x8808 (the only currently defined ctrl packets) are discarded. The support for 802.3Q is optional and need to be enabled via generics.

53.2.3 Clocking

GRETH has three clock domains: The AHB clock, Ethernet receiver clock and the Ethernet transmitter clock. The ethernet transmitter and receiver clocks are generated by the external ethernet PHY, and are inputs to the core through the MII interface. The three clock domains are unrelated to each other and all signals crossing the clock regions are fully synchronized inside the core.

Both full-duplex and half-duplex operating modes are supported and both can be run in either 10 or 100 Mbit. The minimum AHB clock for 10 Mbit operation is 2.5 MHz, while 18 MHz is needed for 100 Mbit. Using a lower AHB clock than specified will lead to excessive packet loss.

53.2.4 RAM debug support

Support for debug accesses the core's internal RAM blocks can be optionally enabled using the ram-debug VHDL generic. Setting it to 1 enables accesses to the transmitter and receiver RAM buffers and setting it to 2 enables accesses to the EDCL buffer in addition to the previous two buffers.

The transmitter RAM buffer is accessed starting from APB address offset 0x10000 which corresponds to location 0 in the RAM. There are 512 32-bit wide locations in the RAM which results in the last address being 0x107FC corresponding to RAM location 511 (byte addressing used on the APB bus).

Correspondingly the receiver RAM buffer is accessed starting from APB address offset 0x20000. The addresses, width and depth is the same.

The EDCL buffers are accessed starting from address 0x30000. The number of locations depend on the configuration and can be from 256 to 16384. Each location is 32-bits wide so the maximum address is 0x3FC and 0xFFFC correspondingly.

Before any debug accesses can be made the ramdebugen bit in the control register has to be set. During this time the debug interface controls the RAM blocks and normal operations is stopped. EDCL packets are not received. The MAC transmitter and receiver could still operate if enabled but the RAM buffers would be corrupt if debug acces are made simultaneously. Thus they MUST be disabled before the RAM debug mode is enabled.

53.2.5 Multibus version

There is a version of the core which has an additional master interface that can be used for the EDCL. Otherwise this version is identical to the basic version. The additional master interface is enabled with the edclsepahb VHDL generic. Then the ethi.edclsepahb signal control whether EDCL accesses are done on the standard master interface or the additional interface. Setting the signal to '0' makes the EDCL use the standard master interface while '1' selects the additional master. This signal is only sampled at reset and changes to this signal have no effect until the next reset.

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53.2.6 Endianness

The core is designed for big-endian systems.

53.3 Tx DMA interface

The transmitter DMA interface is used for transmitting data on an Ethernet network. The transmission is done using descriptors located in memory.

53.3.1 Setting up a descriptor.

A single descriptor is shown in table 787 and 788. The number of bytes to be sent should be set in the length field and the address field should point to the data. The address must be word-aligned. If the interrupt enable (IE) bit is set, an interrupt will be generated when the packet has been sent (this requires that the transmitter interrupt bit in the control register is also set). The interrupt will be generated regardless of whether the packet was transmitted successfully or not. The Wrap (WR) bit is also a control bit that should be set before transmission and it will be explained later in this section.

Table 787. GRETH transmit descriptor word 0 (address offset 0x0)

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31: 16	RESERVED
15	Attempt Limit Error (AL) - The packet was not transmitted because the maximum number of attempts was reached.
14	Underrun Error (UE) - The packet was incorrectly transmitted due to a FIFO underrun error.
13	Interrupt Enable (IE) - Enable Interrupts. An interrupt will be generated when the packet from this descriptor has been sent provided that the transmitter interrupt enable bit in the control register is set. The interrupt is generated regardless if the packet was transmitted successfully or if it terminated with an error.
12	Wrap (WR) - Set to one to make the descriptor pointer wrap to zero after this descriptor has been used. If this bit is not set the pointer will increment by 8. The pointer automatically wraps to zero when the 1 kB boundary of the descriptor table is reached.
11	Enable (EN) - Set to one to enable the descriptor. Should always be set last of all the descriptor fields.
10: 0	LENGTH - The number of bytes to be transmitted.

Table 788. GRETH transmit descriptor word 1 (address offset 0x4)

31	2	1	0
ADDRESS			RES

31: 2	Address (ADDRESS) - Pointer to the buffer area from where the packet data will be loaded.
1: 0	RESERVED

To enable a descriptor the enable (EN) bit should be set and after this is done, the descriptor should not be touched until the enable bit has been cleared by the GRETH.

53.3.2 Starting transmissions

Enabling a descriptor is not enough to start reception. A pointer to the memory area holding the descriptors must first be set in the GRETH. This is done in the receiver descriptor pointer register. The the beginning of the area and must start on an address that is aligned to the size of the descriptor table. The size of the descriptor table can be determined from the formula: STS.NRD*8. The STS.NRD field shows the number of entries in the descriptor table, and each descriptor size is 8 bytes.

Bits 31 to STS.NRD+10 hold the base address of descriptor area while bits STS.NRD+9 to 3 form a pointer to an individual descriptor. The first descriptor should be located at the base address and when it has been used by the GRETH the pointer field is incremented by 8 to point at the next descriptor. The pointer will automatically wrap back to zero when the upper boundary has been reached. The WR bit in the descriptors can be set to make the pointer wrap back to zero before the upper boundary. The pointer field has also been made writable for maximum flexibility but care should be taken when writing to the descriptor pointer register. It should never be touched when a transmission is active.

The final step to activate the transmission is to set the transmit enable bit in the control register. This tells the GRETH that there are more active descriptors in the descriptor table. This bit should always be set when new descriptors are enabled, even if transmissions are already active. The descriptors must always be enabled before the transmit enable bit is set.

53.3.3 Descriptor handling after transmission

When a transmission of a packet has finished, status is written to the first word in the corresponding descriptor. The Underrun Error bit is set if the FIFO became empty before the packet was completely transmitted while the Attempt Limit Error bit is set if more collisions occurred than allowed. The packet was successfully transmitted only if both of these bits are zero. The other bits in the first descriptor word are set to zero after transmission while the second word is left untouched.

The enable bit should be used as the indicator when a descriptor can be used again, which is when it has been cleared by the GRETH. There are three bits in the GRETH status register that hold transmission status. The Transmitter Error (TE) bit is set each time an transmission ended with an error (when at least one of the two status bits in the transmit descriptor has been set). The Transmitter Interrupt (TI) is set each time a transmission ended successfully.

The transmitter AHB error (TA) bit is set when an AHB error was encountered either when reading a descriptor or when reading packet data. Any active transmissions were aborted and the transmitter was disabled. The transmitter can be activated again by setting the transmit enable register.

53.3.4 Setting up the data for transmission

The data to be transmitted should be placed beginning at the address pointed by the descriptor address field. The GRETH does not add the Ethernet address and type fields so they must also be stored in the data buffer. The 4 B Ethernet CRC is automatically appended at the end of each packet. Each descriptor will be sent as a single Ethernet packet. If the size field in a descriptor is greater than defined by maxsize generic + header size bytes, the packet will not be sent.

53.4 Rx DMA interface

The receiver DMA interface is used for receiving data from an Ethernet network. The reception is done using descriptors located in memory.

53.4.1 Setting up descriptors

A single descriptor is shown in table 789 and 790. The address field should point to a word-aligned buffer where the received data should be stored. The GRETH will never store more than defined by the maxsize generic + header size bytes to the buffer. If the interrupt enable (IE) bit is set, an interrupt will be generated when a packet has been received to this buffer (this requires that the receiver interrupt bit in the control register is also set). The interrupt will be generated regardless of whether the packet was received successfully or not. The Wrap (WR) bit is also a control bit that should be set before the descriptor is enabled and it will be explained later in this section.

Table 789. GRETH receive descriptor word 0 (address offset 0x0)

Table 10-10: 10-10 reserved descriptor word 0 (address 00000000)														
31	27	26	25	19	18	17	16	15	14	13	12	11	10	0
RESERVED		MC	RESERVED			LE	OE	CE	FT	AE	IE	WR	EN	LENGTH

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Table 789. GRETH receive descriptor word 0 (address offset 0x0)

31: 27	RESERVED
26	Multicast address (MC) - The destination address of the packet was a multicast address (not broadcast).
25: 19	RESERVED
18	Length error (LE) - The length/type field of the packet did not match the actual number of received bytes.
17	Overrun error (OE) - The frame was incorrectly received due to a FIFO overrun.
16	CRC error (CE) - A CRC error was detected in this frame.
15	Frame too long (FT) - A frame larger than the maximum size was received. The excessive part was truncated.
14	Alignment error (AE) - An odd number of nibbles were received.
13	Interrupt Enable (IE) - Enable Interrupts. An interrupt will be generated when a packet has been received to this descriptor provided that the receiver interrupt enable bit in the control register is set. The interrupt is generated regardless if the packet was received successfully or if it terminated with an error.
12	Wrap (WR) - Set to one to make the descriptor pointer wrap to zero after this descriptor has been used. If this bit is not set the pointer will increment by 8. The pointer automatically wraps to zero when the 1 kB boundary of the descriptor table is reached.
11	Enable (EN) - Set to one to enable the descriptor. Should always be set last of all the descriptor fields.
10: 0	LENGTH - The number of bytes received to this descriptor.

Table 790. GRETH receive descriptor word 1 (address offset 0x4)

31		2	1	0
	ADDRESS			RES
31: 2	Address (ADDRESS) - Pointer to the buffer area from where the packet data will be loaded.			
1: 0	RESERVED			

53.4.2 Starting reception

Enabling a descriptor is not enough to start reception. A pointer to the memory area holding the descriptors must first be set in the GRETH. This is done in the receiver descriptor pointer register. The the beginning of the area and must start on an address that is aligned to the size of the descriptor table. The size of the descriptor table can be determined from the formula: $STS.NRD \times 8$. The STS.NRD field shows the number of entries in the descriptor table, and each descriptor size is 8 bytes.

Bits 31 to STS.NRD+10 hold the base address of descriptor area while bits STS.NRD+9 to 3 form a pointer to an individual descriptor. The first descriptor should be located at the base address and when it has been used by the GRETH the pointer field is incremented by 8 to point at the next descriptor. The pointer will automatically wrap back to zero when the upper boundary has been reached. The WR bit in the descriptors can be set to make the pointer wrap back to zero before the upper boundary.

The pointer field has also been made writable for maximum flexibility but care should be taken when writing to the descriptor pointer register. It should never be touched when reception is active.

The final step to activate reception is to set the receiver enable bit in the control register. This will make the GRETH read the first descriptor and wait for an incoming packet.

53.4.3 Descriptor handling after reception

The GRETH indicates a completed reception by clearing the descriptor enable bit. The other control bits (WR, IE) are also cleared. The number of received bytes is shown in the length field. The parts of the Ethernet frame stored are the destination address, source address, type and data fields. Bits 17-14 in the first descriptor word are status bits indicating different receive errors. All four bits are zero after a reception without errors. The status bits are described in table 789.

Packets arriving that are smaller than the minimum Ethernet size of 64 B are not considered as a reception and are discarded. The current receive descriptor will be left untouched and used for the first packet arriving with an accepted size. The TS bit in the status register is set each time this event occurs.

If a packet is received with an address not accepted by the MAC, the IA status register bit will be set.

Packets larger than maximum size cause the FT bit in the receive descriptor to be set. The length field is not guaranteed to hold the correct value of received bytes. The counting stops after the word containing the last byte up to the maximum size limit has been written to memory.

The address word of the descriptor is never touched by the GRETH.

53.4.4 Reception with AHB errors

If an AHB error occurs during a descriptor read or data store, the Receiver AHB Error (RA) bit in the status register will be set and the receiver is disabled. The current reception is aborted. The receiver can be enabled again by setting the Receive Enable bit in the control register.

53.4.5 Accepted MAC addresses

In the default configuration the core receives packets with either the unicast address set in the MAC address register or the broadcast address. Multicast support can also be enabled and in that case a hash function is used to filter received multicast packets. A 64-bit register, which is accessible through the APB interface, determines which addresses should be received. Each address is mapped to one of the 64 bits using the hash function and if the bit is set to one the packet will be received. The address is mapped to the table by taking the 6 least significant bits of the 32-bit Ethernet crc calculated over the destination address of the MAC frame. A bit in the receive descriptor is set if a packet with a multicast address has been received to it.

53.5 MDIO Interface

The MDIO interface provides access to PHY configuration and status registers through a two-wire interface which is included in the MII interface. The GRETH provided full support for the MDIO interface. If it is not needed in a design it can be removed with a VHDL generic.

The MDIO interface can be used to access from 1 to 32 PHY containing 1 to 32 16-bit registers. A read transfer is set up by writing the PHY and register addresses to the MDIO Control register and setting the read bit. This caused the Busy bit to be set and the operation is finished when the Busy bit is cleared. If the operation was successful the Linkfail bit is zero and the data field contains the read data. An unsuccessful operation is indicated by the Linkfail bit being set. The data field is undefined in this case.

A write operation is started by writing the 16-bit data, PHY address and register address to the MDIO Control register and setting the write bit. The operation is finished when the busy bit is cleared and it was successful if the Linkfail bit is zero.

53.5.1 PHY interrupts

The core also supports status change interrupts from the PHY. A level sensitive interrupt signal can be connected on the mdint input. The mdint_pol vhdL generic can be used to select the polarity. The PHY status change bit in the status register is set each time an event is detected in this signal. If the PHY status interrupt enable bit is set at the time of the event the core will also generate an interrupt on the AHB bus.

53.6 Ethernet Debug Communication Link (EDCL)

The EDCL provides access to an on-chip AHB bus through Ethernet. It uses the UDP, IP and ARP protocols together with a custom application layer protocol. The application layer protocol uses an

ARQ algorithm to provide reliable AHB instruction transfers. Through this link, a read or write transfer can be generated to any address on the AHB bus. The EDCL is optional and must be enabled with a generic.

53.6.1 Operation

The EDCL receives packets in parallel with the MAC receive DMA channel. It uses a separate MAC address which is used for distinguishing EDCL packets from packets destined to the MAC DMA channel. The EDCL also has an IP address which is set through generics. Since ARP packets use the Ethernet broadcast address, the IP-address must be used in this case to distinguish between EDCL ARP packets and those that should go to the DMA-channel. Packets that are determined to be EDCL packets are not processed by the receive DMA channel.

When the packets are checked to be correct, the AHB operation is performed. The operation is performed with the same AHB master interface that the DMA-engines use. The replies are automatically sent by the EDCL transmitter when the operation is finished. It shares the Ethernet transmitter with the transmitter DMA-engine but has higher priority.

53.6.2 EDCL protocols

The EDCL accepts Ethernet frames containing IP or ARP data. ARP is handled according to the protocol specification with no exceptions.

IP packets carry the actual AHB commands. The EDCL expects an Ethernet frame containing IP, UDP and the EDCL specific application layer parts. Table 791 shows the IP packet required by the EDCL. The contents of the different protocol headers can be found in TCP/IP literature.

Table 791. The IP packet expected by the EDCL.

Ethernet Header	IP Header	UDP Header	2 B Offset	4 B Control word	4 B Address	Data 0 - 242 4B Words	Ethernet CRC
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The following is required for successful communication with the EDCL: A correct destination MAC address as set by the generics, an Ethernet type field containing 0x0806 (ARP) or 0x0800 (IP). The IP-address is then compared with the value determined by the generics for a match. The IP-header checksum and identification fields are not checked. There are a few restrictions on the IP-header fields. The version must be four and the header size must be 5 B (no options). The protocol field must always be 0x11 indicating a UDP packet. The length and checksum are the only IP fields changed for the reply.

The EDCL only provides one service at the moment and it is therefore not required to check the UDP port number. The reply will have the original source port number in both the source and destination fields. UDP checksum are not used and the checksum field is set to zero in the replies.

The UDP data field contains the EDCL application protocol fields. Table 792 shows the application protocol fields (data field excluded) in packets received by the EDCL. The 16-bit offset is used to align the rest of the application layer data to word boundaries in memory and can thus be set to any value. The R/W field determines whether a read (0) or a write(1) should be performed. The length

Table 792. The EDCL application layer fields in received frames.

16-bit Offset	14-bit Sequence number	1-bit R/W	10-bit Length	7-bit Unused
---------------	------------------------	-----------	---------------	--------------

field contains the number of bytes to be read or written. If R/W is one the data field shown in table 791 contains the data to be written. If R/W is zero the data field is empty in the received packets. Table 793 shows the application layer fields of the replies from the EDCL. The length field is always zero for replies to write requests. For read requests it contains the number of bytes of data contained in the data field.

Table 793. The EDCL application layer fields in transmitted frames.

16-bit Offset	14-bit sequence number	1-bit ACK/NAK	10-bit Length	7-bit Unused
---------------	------------------------	---------------	---------------	--------------

The EDCL implements a Go-Back-N algorithm providing reliable transfers. The 14-bit sequence number in received packets are checked against an internal counter for a match. If they do not match, no operation is performed and the ACK/NAK field is set to 1 in the reply frame. The reply frame contains the internal counter value in the sequence number field. If the sequence number matches, the operation is performed, the internal counter value is stored in the sequence number field, the ACK/NAK field is set to 0 in the reply and the internal counter is incremented. The length field is always set to 0 for ACK/NAK=1 frames. The unused field is not checked and is copied to the reply. It can thus be set to hold for example some extra identifier bits if needed.

53.6.3 EDCL IP and Ethernet address settings

The default value of the EDCL IP and MAC addresses are set by `ipaddrh`, `ipaddrl`, `macaddrh` and `macaddrl` generics. The IP address and MAC address can later be changed by software. To allow several EDCL enabled GRETH controllers on the same sub-net, the 4 LSB bits of the IP and MAC address can optionally be set by an input signal. This is enabled by setting the `edcl` generic = 2, and driving the 4-bit LSB value on `ethi.edcladdr`.

53.6.4 EDCL buffer size

The EDCL has a dedicated internal buffer memory which stores the received packets during processing. The size of this buffer is configurable with a VHDL generic to be able to obtain a suitable compromise between throughput and resource utilization in the hardware. Table 794 lists the different buffer configurations. For each size the table shows how many concurrent packets the EDCL can handle, the maximum size of each packet including headers and the maximum size of the data payload. Sending more packets before receiving a reply than specified for the selected buffer size will lead to dropped packets. The behavior is unspecified if sending larger packets than the maximum allowed.

Table 794. EDCL buffer sizes

Total buffer size (kB)	Number of packet buffers	Packet buffer size (B)	Maximum data payload (B)
1	4	256	200
2	4	512	456
4	8	512	456
8	8	1024	968
16	16	1024	968
32	32	1024	968
64	64	1024	968

53.7 Media Independent Interfaces

There are several interfaces defined between the MAC sublayer and the Physical layer. The GRETH supports two of them: The Media Independent Interface (MII) and the Reduced Media Independent Interface (RMII).

The MII was defined in the 802.3 standard and is most commonly supported. The ethernet interface have been implemented according to this specification. It uses 16 signals.

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To support lower speed where the operation and clock frequency of the core and phy remains unchanged i.e. running at 10Mb/s when the IP is configured for 100Mb/s speed enable signals should be created to mimic the desired bit rate.

When operating at 10Mb/s, every byte of the MAC frame is repeated 10 clock periods to achieve the correct bit rate. The GRETH_GBIT core does not take care of this operation and enable signals with toggling frequency of the correct bit rate needs to be created.

The RMII was developed to meet the need for an interface allowing Ethernet controllers with smaller pin counts. It uses 6 (7) signals which are a subset of the MII signals. Table 795 shows the mapping between the RMII signals and the GRLIB MII interface.

Table 795. Signal mappings between RMII and the GRLIB MII interface.

RMII	MII
txd[1:0]	txd[1:0]
tx_en	tx_en
crs_dv	rx_crs
rx_d[1:0]	rx_d[1:0]
ref_clk	rmii_clk
rx_er	not used

53.8 AXI support

The core is designed for an AMBA system but can be adapted for AXI using the AHBM2AXI adapter.

53.9 Registers

The core is programmed through registers mapped into APB address space.

Table 796. GRETH registers

APB address offset	Register
0x0	Control register
0x4	Status/Interrupt-source register
0x8	MAC Address MSB
0xC	MAC Address LSB
0x10	MDIO Control/Status
0x14	Transmit descriptor pointer
0x18	Receiver descriptor pointer
0x1C	EDCL IP
0x20	Hash table msb
0x24	Hash table lsb
0x28	EDCL MAC address MSB
0x2C	EDCL MAC address LSB
0x10000 - 0x107FC	Transmit RAM buffer debug access
0x20000 - 0x207FC	Receiver RAM buffer debug access
0x30000 - 0x3FFFC	EDCL buffer debug access

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53.9.1 Control Register

Table 797.0x00 - CTRL - GRETH control register

31	30	28	27	26	25	24	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EA	BS			MA	MC	RESERVED	ED	RD	DD	ME	PI	RES	SP	RS	PM	FD	RI	TI	RE	TE		
*	*	*	*	*		0	*	*	0	0	0	0	0	1	0	0	0	0	0	0	0	0
r	r	r	r	r		r	rw	rw	rw	rw	rw	r	rw	wc	rw	rw	rw	rw	rw	rw	rw	rw

- 31 EDCL available (EA) - Set to one if the EDCL is available.
- 30: 28 EDCL buffer size (BS) - Shows the amount of memory used for EDCL buffers. 0 = 1 kB, 1 = 2 kB, ..., 6 = 64 kB.
- 27 RESERVED
- 26 MDIO interrupts available (MA) - Set to one when the core supports mdio interrupts. Read only.
- 25 Multicast available (MC) - Set to one when the core supports multicast address reception. Read only.
- 24: 15 RESERVED
- 14 EDCL Disable (ED) - Set to one to disable the EDCL and zero to enable it. Reset value taken from the ethi.edcldisable signal. Only available if the EDCL hardware is present in the core.
- 13 RAM debug enable (RD) - Set to one to enable the RAM debug mode. Reset value: '0'. Only available if the VHDL generic ramdebug is nonzero.
- 12 Disable duplex detection (DD) - Disable the EDCL speed/duplex detection FSM. If the FSM cannot complete the detection the MDIO interface will be locked in busy mode. If software needs to access the MDIO the FSM can be disabled here and as soon as the MDIO busy bit is 0 the interface is available. Note that the FSM cannot be reenabled again.
- 11 Multicast enable (ME) - Enable reception of multicast addresses. Reset value: '0'.
- 10 PHY status change interrupt enable (PI) - Enables interrupts for detected PHY status changes.
- 9: 8 RESERVED
- 7 Speed (SP) - Sets the current speed mode. 0 = 10 Mbit, 1 = 100 Mbit. Only used in RMII mode (rmii = 1). A default value is automatically read from the PHY after reset. Reset value: '1'.
- 6 Reset (RS) - A one written to this bit resets the GRETH core. Self clearing. No other accesses should be done to the slave interface other than polling this bit until it is cleared.
- 5 Promiscuous mode (PM) - If set, the GRETH operates in promiscuous mode which means it will receive all packets regardless of the destination address. Reset value: '0'.
- 4 Full duplex (FD) - If set, the GRETH operates in full-duplex mode otherwise it operates in half-duplex. Reset value: '0'.
- 3 Receiver interrupt (RI) - Enable Receiver Interrupts. An interrupt will be generated each time a packet is received when this bit is set. The interrupt is generated regardless if the packet was received successfully or if it terminated with an error. Reset value: '0'.
- 2 Transmitter interrupt (TI) - Enable Transmitter Interrupts. An interrupt will be generated each time a packet is transmitted when this bit is set. The interrupt is generated regardless if the packet was transmitted successfully or if it terminated with an error. Reset value: '0'.
- 1 Receive enable (RE) - Should be written with a one each time new descriptors are enabled. As long as this bit is one the GRETH will read new descriptors and as soon as it encounters a disabled descriptor it will stop until RE is set again. This bit should be written with a one after the new descriptors have been enabled. Reset value: '0'.
- 0 Transmit enable (TE) - Should be written with a one each time new descriptors are enabled. As long as this bit is one the GRETH will read new descriptors and as soon as it encounters a disabled descriptor it will stop until TE is set again. This bit should be written with a one after the new descriptors have been enabled. Reset value: '0'.

53.9.2 Status Register

Table 798.0x04 - STAT - GRETH status register

31	28	27	24	23	9	8	7	6	5	4	3	2	1	0
RESERVED		NRD		RESERVED	PS	IA	TS	TA	RA	TI	RI	TE	RE	
0		*		0	0	0	0	NR	NR	NR	NR	NR	NR	
r		r		r	wc	wc	wc	wc	wc	wc	wc	wc	wc	

Table 798.0x04 - STAT - GRETH status register

27: 24	Number of receive descriptors (NRD) - Shows the size of the DMA descriptor table. 0b0000 = 128, 0b0001 = 256, 0b0010 = 512, 0b0011 = 1024, 0b0100 = 2048, 0b0101 = 4096, 0b0110 = 8192, 0b0111 = 16384, 0b1000 = 32768, 0b1001 = 65536
8	PHY status changes (PS) - Set each time a PHY status change is detected.
7	Invalid address (IA) - A packet with an address not accepted by the MAC was received. Cleared when written with a one. Reset value: '0'.
6	Too small (TS) - A packet smaller than the minimum size was received. Cleared when written with a one. Reset value: '0'.
5	Transmitter AHB error (TA) - An AHB error was encountered in transmitter DMA engine. Cleared when written with a one. Not Reset.
4	Receiver AHB error (RA) - An AHB error was encountered in receiver DMA engine. Cleared when written with a one. Not Reset.
3	Transmitter interrupt (TI) - A packet was transmitted without errors. Cleared when written with a one. Not Reset.
2	Receiver interrupt (RI) - A packet was received without errors. Cleared when written with a one. Not Reset.
1	Transmitter error (TE) - A packet was transmitted which terminated with an error. Cleared when written with a one. Not Reset.
0	Receiver error (RE) - A packet has been received which terminated with an error. Cleared when written with a one. Not Reset.

53.9.3 MAC Address MSB

Table 799.0x08 - MACMSB - GRETH MAC address MSB.

31	16	15	0
RESERVED		Bit 47 downto 32 of the MAC address	
		NR	
		rw	

31: 16 RESERVED

15: 0 The two most significant bytes of the MAC Address. Not Reset.

53.9.4 MAC Address LSB

Table 800.0x0C - MACLSB - GRETH MAC address LSB.

31	0
Bit 31 downto 0 of the MAC address	
NR	
rw	

31: 0 The four least significant bytes of the MAC Address. Not Reset.

53.9.5 MDIO ctrl/status Register

Table 801.0x10 - MDIO - GRETH MDIO ctrl/status register.

31	16	15	11	10	6	5	4	3	2	1	0
DATA		PHYADDR	REGADDR		RES	BU	LF	RD	WR		
0		*	0			0	1	0	0		
rw		rw	rw			r	r	rw	rw		

31: 16 Data (DATA) - Contains data read during a read operation and data that is transmitted is taken from this field. Reset value: 0x0000.

15: 11 PHY address (PHYADDR) - This field contains the address of the PHY that should be accessed during a write or read operation. Reset value: "00000".

10: 6 Register address (REGADDR) - This field contains the address of the register that should be accessed during a write or read operation. Reset value: "00000".

5:4 RESERVED

3 Busy (BU) - When an operation is performed this bit is set to one. As soon as the operation is finished and the management link is idle this bit is cleared. Reset value: '0'.

2 Linkfail (LF) - When an operation completes (BUSY = 0) this bit is set if a functional management link was not detected. Reset value: '1'.

1 Read (RD) - Start a read operation on the management interface. Data is stored in the data field. Reset value: '0'.

0 Write (WR) - Start a write operation on the management interface. Data is taken from the Data field. Reset value: '0'.

53.9.6 Transmitter Descriptor Table Base Address Register

Table 802.0x14 - TXBASE - GRETH transmitter descriptor table base address register.

31	X+1	X	3	2	0
BASEADDR		DESCPNT	RES		
NR		0	0		
rw		rw	r		

- 31: X+1 Transmitter descriptor table base address (BASEADDR) - Base address to the transmitter descriptor table. Not Reset. The value of x is given by the formula: $9 + \text{STS.NXD}$
- X: 3 Descriptor pointer (DESCPNT) - Pointer to individual descriptors. Automatically incremented by the Ethernet MAC. The value of x is given by the formula: $9 + \text{STS.NXD}$.
- 2: 0 RESERVED

53.9.7 Receiver Descriptor Table Base Address Register

Table 803.0x18 - RXBASE - GRETH receiver descriptor table base address register.

31	X+1	X	3	2	0
BASEADDR		DESCPNT	RES		
NR		0	0		
rw		rw	r		

- 31: X+1 Receiver descriptor table base address (BASEADDR) - Base address to the receiver descriptor table. Not Reset. The value of x is given by the formula: $9 + \text{STS.NXD}$
- X: 3 Descriptor pointer (DESCPNT) - Pointer to individual descriptors. Automatically incremented by the Ethernet MAC. The value of x is given by the formula: $9 + \text{STS.NXD}$
- 2: 0 RESERVED

53.9.8 EDCL IP Register

Table 804.0x1C - EDCLIP - GRETH EDCL IP register

31	0
EDCL IP ADDRESS	
*	
rw	

- 31: 0 EDCL IP address. Reset value is set with the ipaddrh and ipaddrl generics.

53.9.9 Hash Table Msb Register

Table 805.0x20 - HhSB - GRETH Hash table msb register

31	0
Hash table (64:32)	
NR	
rw	

- 31: 0 Hash table msb. Bits 64 downto 32 of the hash table.

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53.9.10 Hash Table Lsb Register

Table 806.0x24 - HCSB - GRETH Hash table lsb register

31	0
Hash table (64:32)	
NR	
rw	

31: 0 Hash table lsb. Bits 31 downto 0 of the hash table.

53.9.11 EDCL MAC Address MSB

Table 807.0x28 - EMACMSB - GRETH EDCL MAC address MSB.

31	16	15	0
RESERVED		Bit 47 downto 32 of the EDCL MAC Address	
		*	
		rw	

31: 16 RESERVED

15: 0 The two most significant bytes of the EDCL MAC Address. Hardcoded reset value set with the VHDL generic macaddrh.

53.9.12 EDCL MAC Address LSB

Table 808.0x2C - EMACLSB - GRETH EDCL MAC address LSB.

31	0
Bit 31 downto 0 of the EDCL MAC Address	
*	
rw	

31: 0 The 4 least significant bytes of the EDCL MAC Address. Hardcoded reset value set with the VHDL generics macaddrh and macaddrl. If the VHDL generic edcl is set to 2 bits 3 downto 0 are set with the edcladdr input signal.

53.10 Vendor and device identifiers

The core has vendor identifier 0x01 (Frontgrade Gaisler) and device identifier 0x1D. For description of vendor and device identifiers see GRLIB IP Library User's Manual.

53.11 Implementation

53.11.1 Reset

The core does not change reset behaviour depending on settings in the GRLIB configuration package (see GRLIB User's Manual). The core makes use of synchronous reset and resets a subset of its internal registers.

53.11.2 Endianness

The core changes endianness behaviour depending on the settings in the GRLIB configuration package (see GRLIB User's Manual). Changing *grib_little_endian* controls TBD